

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

ACCIDENT ANALYTICS AND VISUALIZATION USING POWER BI

Framework for Data Visual Analytics

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ABSTRACT

This mini project, carried out for the subject Framework for Data Visual Analytics (FDVA), studies road accident risk using Microsoft Power BI. The dataset has 350 accident records and 16 columns for five districts of Tamil Nadu (Chennai, Coimbatore, Madurai, Tirunelveli and Vellore) from January 2024 to December 2025. It was created in Python with Pandas and NumPy, and it describes each accident by district, location type, road type, road condition, weather, time slot, vehicle type, driver age group, vehicles involved, fatalities, injuries, severity, speed limit and main cause. The data was checked in Python and had no missing values and no duplicate rows. Instead of only counting accidents, the analysis focuses on how risky each condition is: the share of fatal accidents was measured for each district, vehicle type, speed limit and weather condition. Twelve visuals were prepared in Power BI style: a combo chart, a Pareto chart, a 100% stacked bar chart, a line chart with a moving average, two matrices, a donut chart, a box and whisker chart, a stacked column chart, a bubble scatter plot and a column chart. The analysis showed that 42.9% of the accidents were fatal, that two-wheelers had the highest share of fatal accidents (55.4%), and that the share of fatal accidents rose steadily with the speed limit, from 11% at 30 km/h to 82% at 100 km/h. Overspeeding, distracted driving and drunk driving together caused 51% of accidents, and the largest share of accidents happened between 4 pm and 8 pm. The project shows how a risk-based view of simple charts can support road safety decisions.

1. Dataset and Data Preparation

The dataset has 350 road accident records and 16 columns. It covers five districts and the period from January 2024 to December 2025. The data was created in Python using Pandas and NumPy (random seed 2025) and was checked before the charts were made. Because the data is simulated, the results describe this dataset only and are not real accident statistics.

1.1 Fields in the dataset

Each row is one accident. The main fields are listed below.

- **Accident ID, Date and District:** identifier, date and district of the accident (Chennai, Coimbatore, Madurai, Tirunelveli or Vellore).
- **Location type, Road type and Road condition:** Urban, Suburban or Rural; National Highway, State Highway, City Road or District Road; and Normal, Wet, Potholes or Under Construction.
- **Weather condition and Time slot:** Clear, Raining or Mist / Fog; and one of six four-hour slots of the day (00-04, 04-08, 08-12, 12-16, 16-20 and 20-24).
- **Vehicle type, Driver age group and Vehicles involved:** Two-wheeler, Car, Auto-rickshaw, Bus, Truck or Lorry; the age band of the driver (18-25 to 56+); and the number of vehicles in the accident.
- **Fatalities, Injuries and Accident severity:** number of deaths and injuries, and the severity (Fatal if at least one death, Moderate if two or more injuries, otherwise Minor).
- **Speed limit and Main cause of accident:** speed limit of the road in km/h and the main reason for the accident (eight causes).

1.2 Checking the data

```
print(df.info())
print(df.isnull().sum())
print(df.duplicated().sum())
print(df.shape)   # (350, 16)
```

The data frame has 350 entries and 16 columns (4 numeric, 1 date and 11 text). Every column has 350 non-null values, so there are no missing values and no duplicate rows, and no further cleaning was needed.

1.3 Loading the data into Power BI

```
df.to_csv("accident_data.csv", index=False)
```

The data was saved as a CSV file and loaded into Power BI with Get data, and one visual was placed on each of twelve report pages. Counts use Count of Accident ID. Power BI has

no built-in Pareto chart, heat map or bubble-size chart, so a column and line chart, a matrix with conditional formatting and a scatter chart with the Size field are used instead. The Day of week and Month fields were derived from the Date column.

1.4 Derived measures

To compare risk and not only volume, two measures were added. The fatal accident rate is the number of fatal accidents divided by all accidents in the same group, and casualties are fatalities plus injuries.

```
Fatal % = DIVIDE(
    CALCULATE(COUNTROWS(accident_data),
              accident_data[Accident severity] = "Fatal"),
    COUNTROWS(accident_data))
Casualties = SUM(accident_data[Fatalities])
           + SUM(accident_data[Injuries])
```

2. Data Visualization Using Power BI

Twelve visuals were prepared to study accidents by district, cause, vehicle, month, day and time, road type, injuries, weather, speed limit, driver age, road condition and quarter. Each chart is explained briefly below.

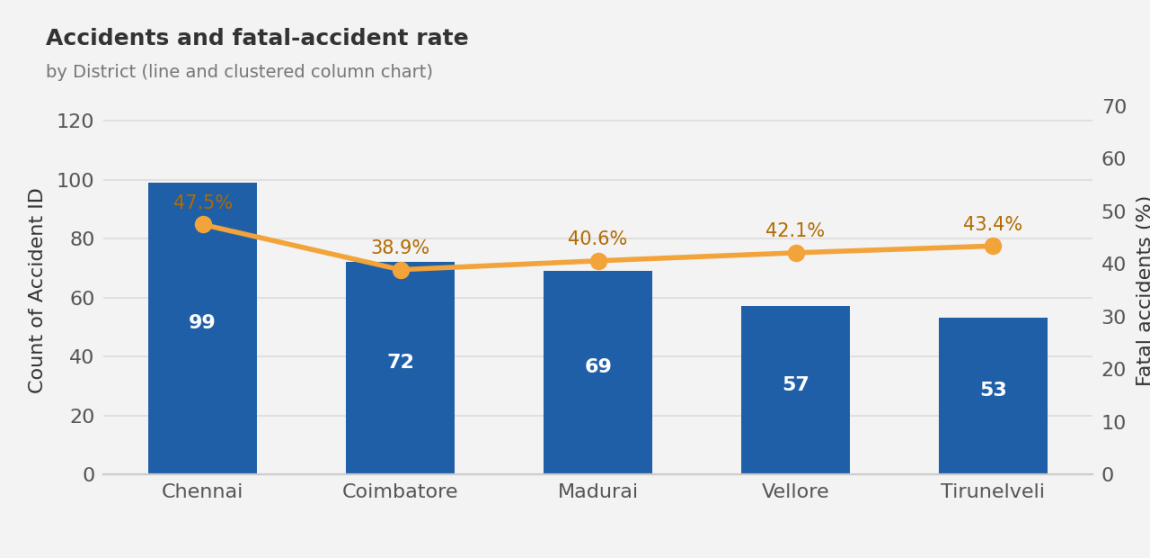


Figure 2.1: Accidents and fatal-accident rate by district (combo chart)

Result: Chennai has the most accidents (99, about 28% of 350), followed by Coimbatore (72) and Madurai (69). Tirunelveli has the fewest (53). Chennai also has the highest fatal accident rate (47.5%), while Coimbatore has the lowest (38.9%), so the busiest district is also the riskiest.

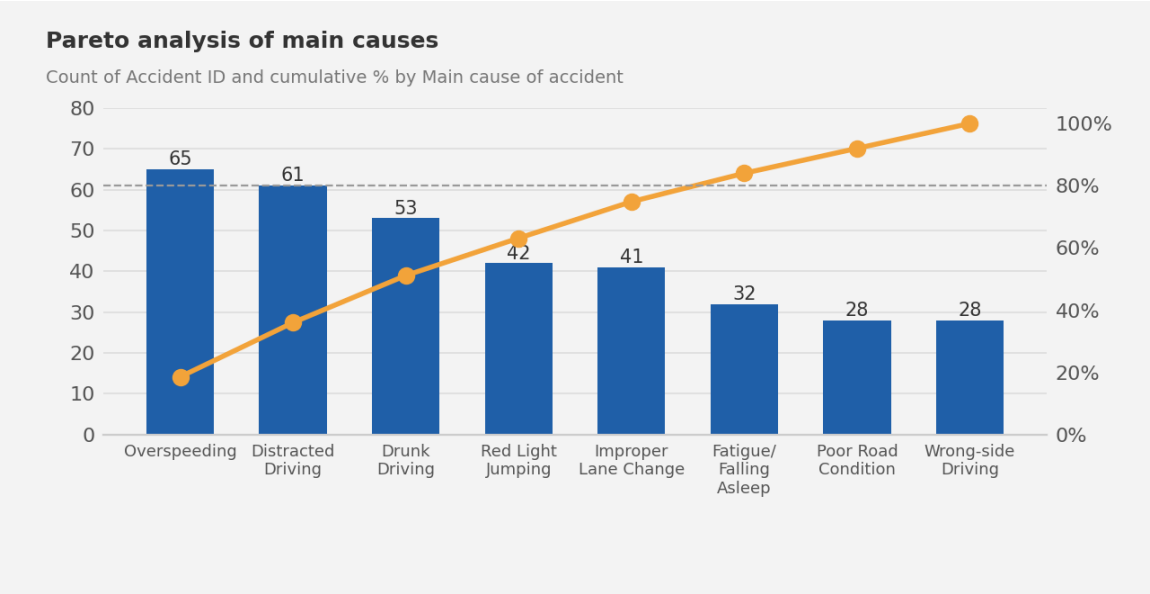


Figure 2.2: Pareto chart of main causes of accidents

Result: Overspeeding (65), distracted driving (61) and drunk driving (53) together account for 51.1% of accidents, and the top five causes reach 74.9%. Poor road condition and wrong-side driving are the lowest (28 each). A few driver behaviours explain most accidents.

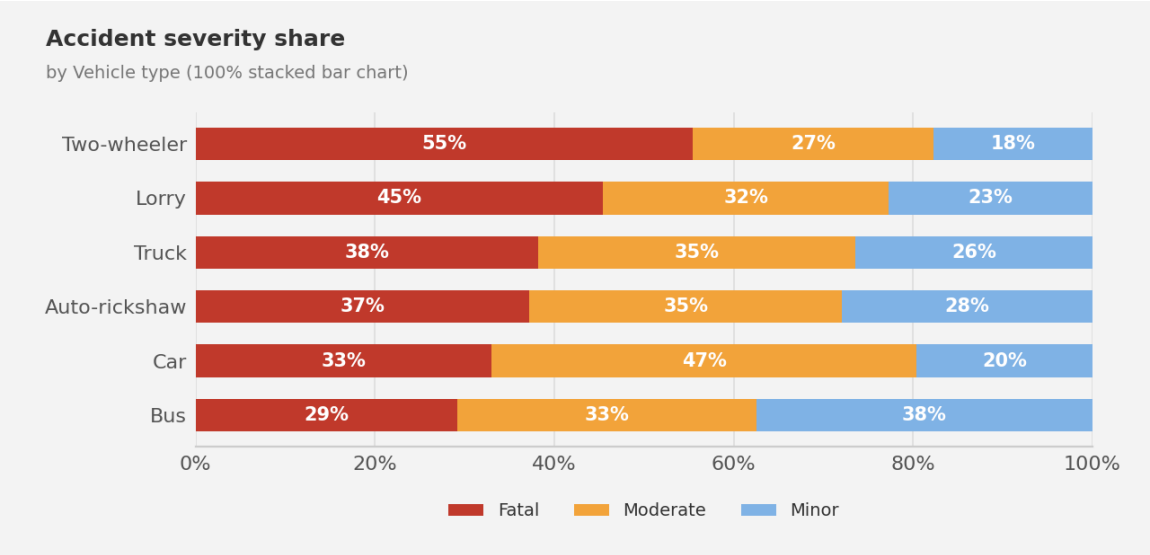


Figure 2.3: Accident severity share by vehicle type (100% stacked bar chart)

Result: Two-wheelers have the highest share of fatal accidents (55%), followed by lorries (45%). Buses have the lowest (29%) and also the largest share of minor accidents (38%). Cars and auto-rickshaws are close at 33% and 37%.

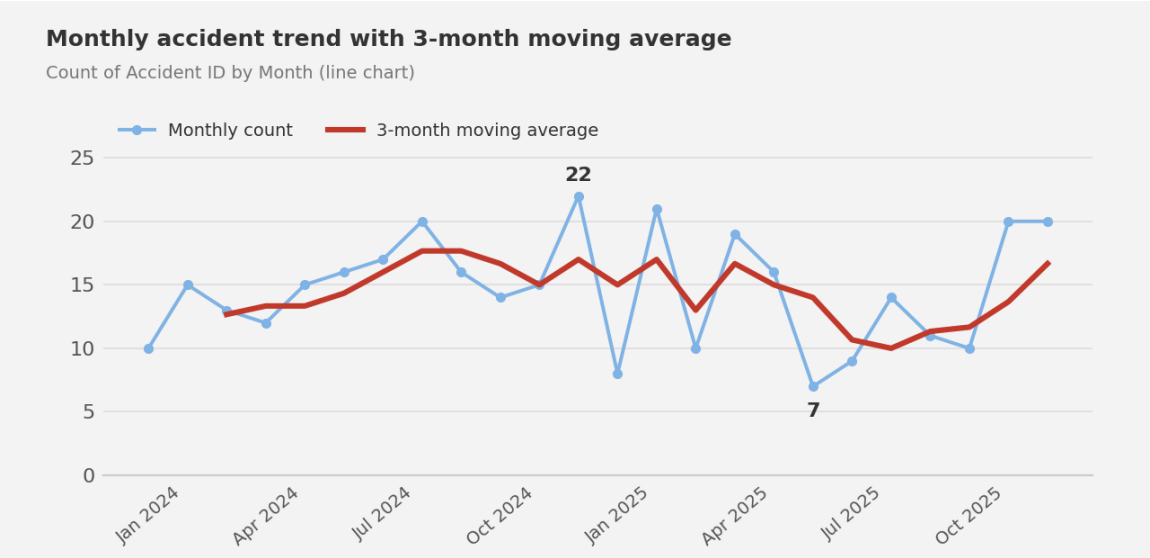


Figure 2.4: Monthly accident trend with 3-month moving average (line chart)

Result: Monthly accidents vary between 7 and 22. The highest month is December 2024 (22) and the lowest is June 2025 (7). The moving average falls through the middle of 2025 and rises again in the last months. The year 2024 had 185 accidents and 2025 had 165, a fall of about 11%.

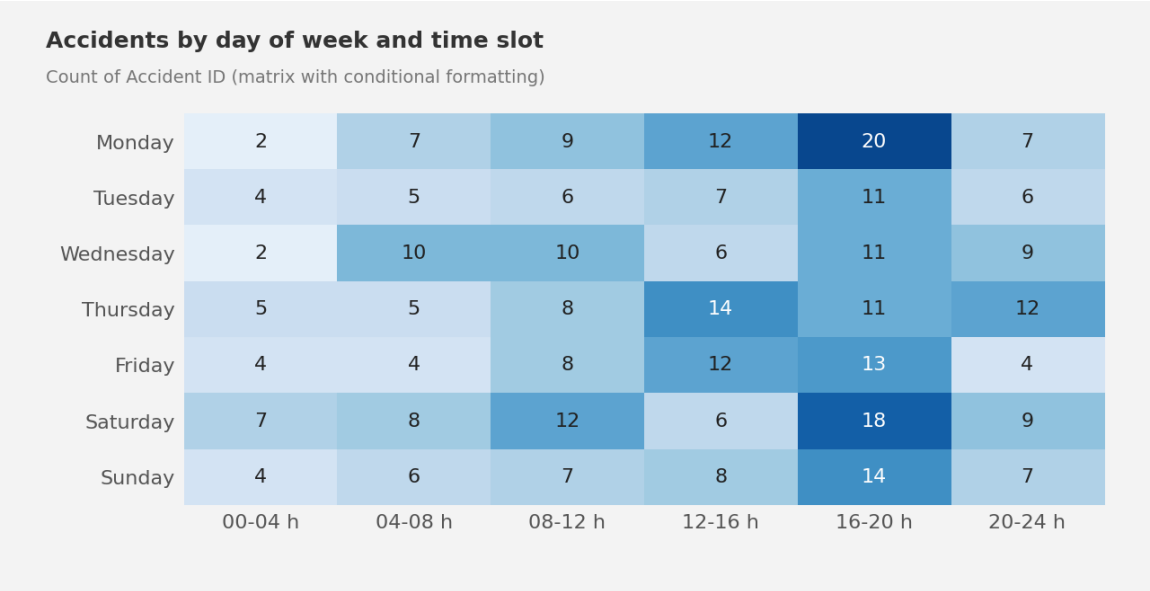


Figure 2.5: Accidents by day of week and time slot (matrix heat map)

Result: The 16-20 h slot is the busiest, with 98 accidents (28%), and the single highest cell is Monday evening (20). The night slot 00-04 h is the quietest (28). Saturday has the most accidents (60) and Tuesday the fewest (39).

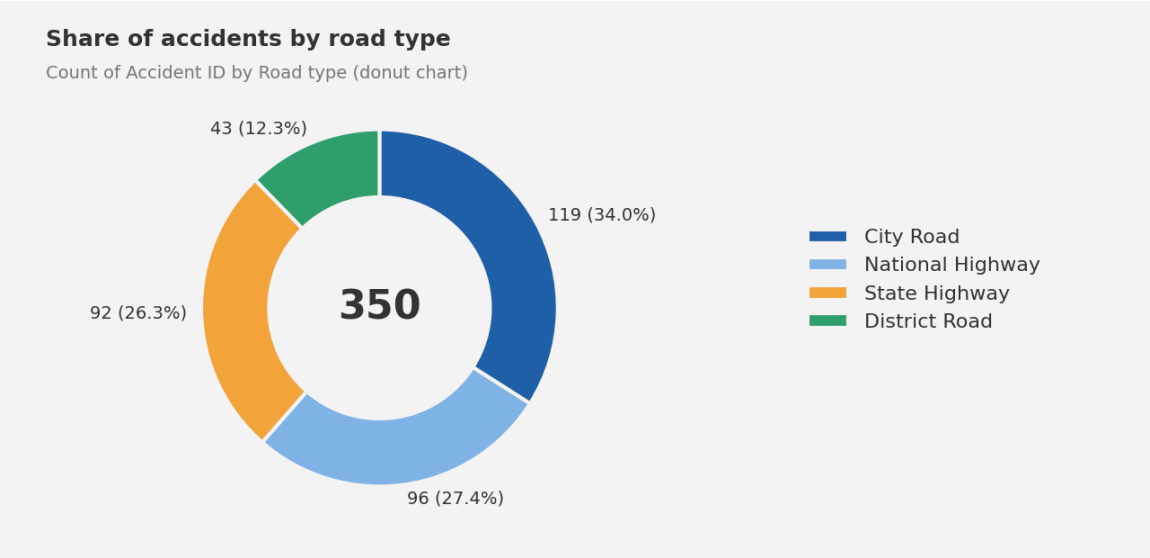


Figure 2.6: Share of accidents by road type (donut chart)

Result: City roads have the largest share (119, 34.0%), followed by national highways (96, 27.4%) and state highways (92, 26.3%). District roads have the smallest share (43, 12.3%).

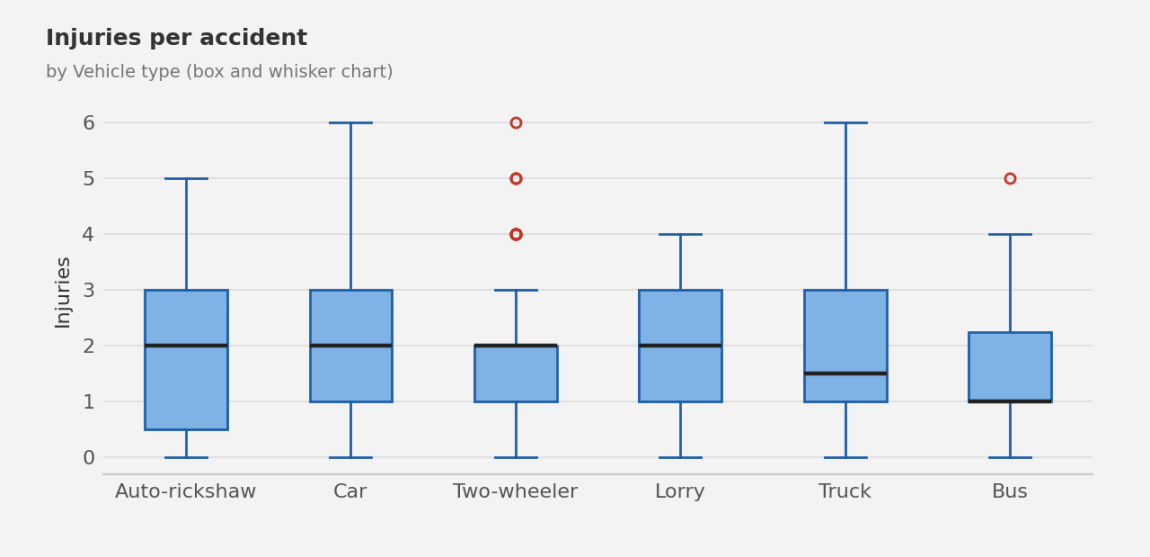


Figure 2.7: Injuries per accident by vehicle type (box and whisker chart)

Result: The median number of injuries is 2 for most vehicle types, 1.5 for trucks and 1 for buses. Cars, trucks and two-wheelers reach 6 injuries in some accidents, and the spread of injuries is similar across vehicle types.

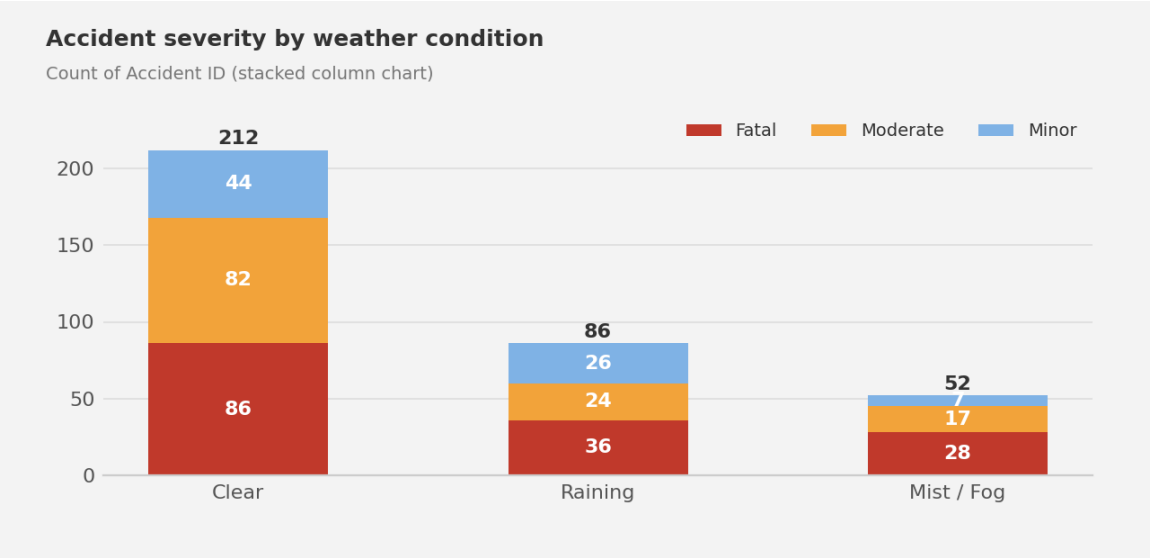


Figure 2.8: Accident severity by weather condition (stacked column chart)

Result: Clear weather has the most accidents (212, about 61%), then rain (86) and mist / fog (52). However, mist / fog has the highest share of fatal accidents (28 of 52, 53.8%) compared with clear weather (40.6%) and rain (41.9%).

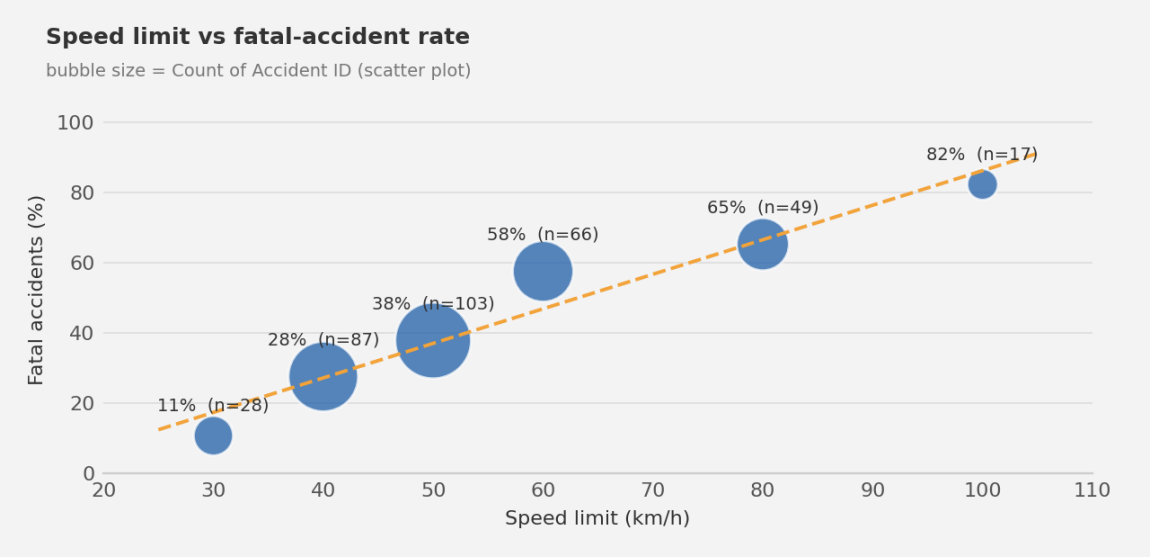


Figure 2.9: Speed limit vs fatal-accident rate (bubble scatter plot)

Result: The share of fatal accidents rises steadily with the speed limit: 11% at 30 km/h, 38% at 50 km/h, 58% at 60 km/h, 65% at 80 km/h and 82% at 100 km/h. Most accidents are on 40 and 50 km/h roads, but the faster roads are far more deadly.

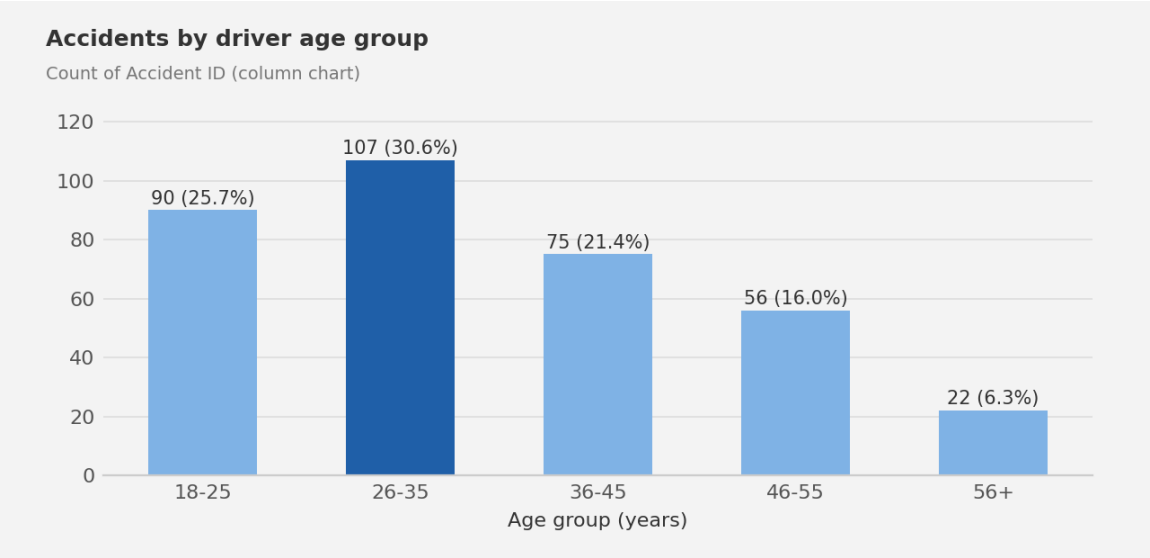


Figure 2.10: Accidents by driver age group (column chart)

Result: Drivers aged 26-35 are involved in the most accidents (107, 30.6%), followed by 18-25 (90, 25.7%). These two groups together make up 56% of accidents, while drivers aged 56 and above have the fewest (22, 6.3%).

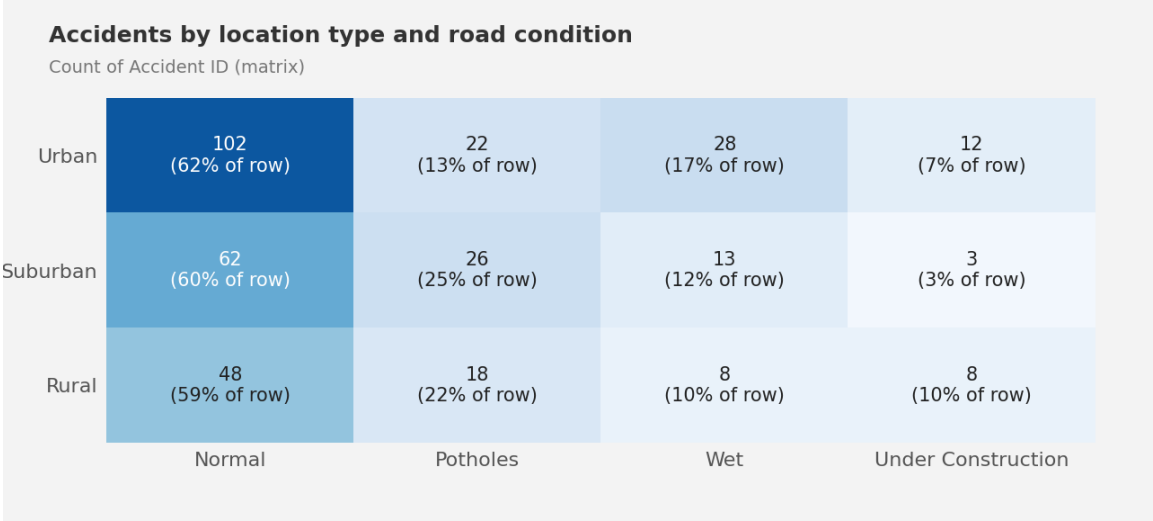


Figure 2.11: Accidents by location type and road condition (matrix)

Result: Normal roads carry about 60% of accidents in every location type. Potholes make up a larger share in suburban (25%) and rural (22%) areas than in urban areas (13%), and wet roads are most common in urban areas (17%).

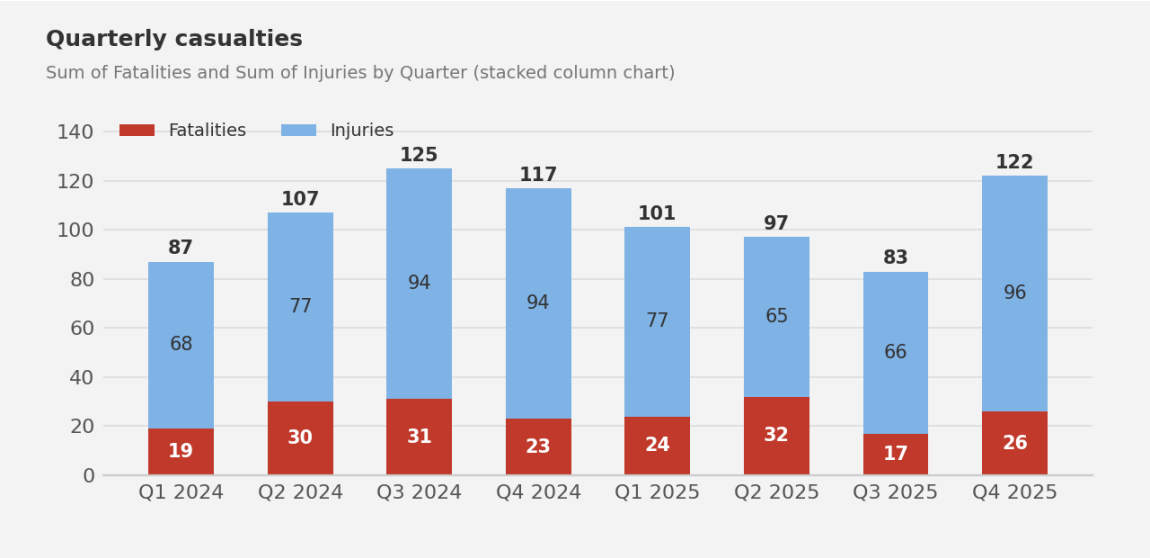


Figure 2.12: Quarterly casualties (stacked column chart)

Result: The data has 202 deaths and 637 injuries, which is 839 casualties or 2.4 per accident. Casualties were highest in Q3 2024 (125) and lowest in Q3 2025 (83). Deaths peaked in Q2 2025 (32), and injuries peaked in Q4 2025 (96).

Overall, the busiest conditions are Chennai, two-wheelers, city roads, clear weather and the evening hours, but the most dangerous conditions are high speed limits, two-wheelers and mist / fog.

3. Key Findings

Comparing the charts together gives the following points.

- **Where:** Chennai has the highest share of accidents (28%) and the highest fatal accident rate (47.5%); Tirunelveli has the lowest share (about 15%).
- **Who:** two-wheelers and cars together are involved in about 65% of accidents, and drivers aged 18 to 35 in about 56%. Two-wheelers are fatal in more than half of their accidents (55.4%).
- **Severity:** 42.9% of accidents are fatal and 22.0% are minor, with 2.4 casualties per accident.
- **Speed:** the fatal accident rate rises with every step of the speed limit, from 11% at 30 km/h to 82% at 100 km/h.
- **Causes:** overspeeding, distracted driving and drunk driving cause 51% of accidents, and poor road condition is the main cause in only 28 of 350.
- **Time:** 28% of accidents happen between 4 pm and 8 pm and the weekday peak is Monday evening; monthly counts change a lot (7 to 22) and fell by about 11% from 2024 to 2025.
- **Conditions:** most accidents are in clear weather and normal road condition, so these counts show where accidents were recorded and not which conditions are riskier. Looking at the fatal accident rate instead shows that mist / fog is the riskiest weather (53.8%).

4. Conclusion

This mini project used Power BI-style visuals to analyse 350 road accident records from five districts. The analysis compared the risk of each condition and not only the number of accidents. The charts show that Chennai, two-wheelers, city roads and the evening hours have the most recorded accidents, that overspeeding, distracted driving and drunk driving are the leading causes, and that the share of fatal accidents rises sharply with the speed limit and is highest for two-wheelers and in mist / fog. These results can help road safety teams focus on speed control, helmet and two-wheeler safety, drunk-driving checks and evening patrols in the busiest districts. Because the data is simulated, the same steps should be repeated on real accident records before drawing real-world conclusions.